

# Package ‘EMGCR’

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**Type** Package

**Title** Fit a Mixture Cure Rate Model With Custom Link Function

**Version** 0.1.0

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**Description** Tools to fit Mixture Cure Rate (MCR) models via the Expectation-Maximization (EM) algorithm, allowing for flexible link functions in the cure component and various survival distributions in the latency part. The package supports user-specified link functions, includes methods for parameter estimation and model diagnostics, and provides residual analysis tailored for cure models.

**License** GPL-3

**Encoding** UTF-8

**Depends** R (>= 4.2.0)

**Imports** survival,  
Formula,  
knitr,  
actuar,  
flexsurv,  
tibble,  
ggplot2

**Roxygen** list(markdown = TRUE)

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liver

*Liver Cancer Data*

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## Description

A sample of 2,766 patients diagnosed with liver cancer between 2012 and 2016, whose cancer grades were well identified. Available individual-level covariates include age at diagnosis, sex, histological grade of the liver cancer, number of relapses, and median household income.

## Usage

```
data(liver)
```

## Format

A data frame with 2,766 observations and 10 variables:

**ID** Unique patient identifier

**age** Age as a factor (e.g., age group)

**ageNumeric** Age as a numeric variable

**grade** Histologic grade of liver cancer (I, II, III, IV)

**medh** Median household income as factor (possibly grouped)

**medhNumeric** Median household income as numeric

**relapse** Number of relapses after first diagnosis

**sex** Sex of patient: 1 = male, 0 = female

**status** Event indicator: 1 = death, 0 = censored

**time** Survival time in months

## Details

The grade of the disease is categorized into four levels:

1. Grade I – Well differentiated
2. Grade II – Moderately differentiated
3. Grade III – Poorly differentiated
4. Grade IV – Undifferentiated/anaplastic

## Examples

```
data(liver)
head(liver)
```

MCRfit

*Fit a Mixture Cure Rate (MCR) Survival Model***Description**

Fits a cure rate model using a flexible link function and a variety of survival distributions. The model accounts for a cured fraction through a logistic-type link and estimates the model via an EM-like algorithm.

**Usage**

```
MCRfit(
  formula,
  data,
  dist = "weibull",
  link = "logit",
  tau = 1,
  maxit = 1000,
  tol = 1e-05
)
```

**Arguments**

|         |                                                                                                                                                                                         |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| formula | A two-part formula of the form $\text{Surv}(\text{time}, \text{status}) \sim x \mid w$ , where $x$ are co-variates for the survival part, and $w$ are covariates for the cure fraction. |
| data    | A data frame containing the variables in the model.                                                                                                                                     |
| dist    | A character string indicating the baseline distribution. Supported values are "weibull", "exponential", "rayleigh", "lognormal", "loglogistic", and "invgauss".                         |
| link    | A character string specifying the link function for the cure fraction. Options are "logit", "probit", "plogit", "rplogit", and "cauchit".                                               |
| tau     | A numeric value used when link = "plogit" or "rplogit". Defaults to 1.                                                                                                                  |
| maxit   | Maximum number of iterations for the EM-like algorithm. Defaults to 1000.                                                                                                               |
| tol     | Convergence tolerance. Defaults to 1e-5.                                                                                                                                                |

**Value**

An object of class "MCR", which is a list containing:

|                   |                                                          |
|-------------------|----------------------------------------------------------|
| coefficients      | Estimated regression coefficients for the survival part. |
| coefficients_cure | Estimated coefficients for the cure part.                |
| scale             | Estimated scale parameter of the baseline distribution.  |
| loglik            | Final log-likelihood value.                              |
| n                 | Number of observations used in the model.                |
| deleted           | Number of incomplete cases removed before fitting.       |
| ep                | Estimated standard errors.                               |
| iter              | Number of iterations used for convergence.               |

|      |                                     |
|------|-------------------------------------|
| dist | Distribution used.                  |
| link | Link function used.                 |
| tau  | Tau parameter used (if applicable). |

### Examples

```
require(EMGCR)

data(liver)
names(liver)

model <- MCRfit(
  survival::Surv(time, status) ~ age + medh + relapse + grade | sex + age + medh + grade,
  data = liver
)
summary(model)
```

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|          |                                                                |
|----------|----------------------------------------------------------------|
| plot.MCR | <i>Plot multiple MCR model fits against Kaplan-Meier curve</i> |
|----------|----------------------------------------------------------------|

---

### Description

Plot multiple MCR model fits against Kaplan-Meier curve

### Usage

```
## S3 method for class 'MCR'
plot(...)
```

### Arguments

... One or more fitted MCR objects from `MCRfit()`.

### Value

A ggplot object with Kaplan-Meier and survival curves for each model.

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|       |                                           |
|-------|-------------------------------------------|
| qqMCR | <i>QQ-Plot of Residuals for MCR Model</i> |
|-------|-------------------------------------------|

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### Description

Produces a Q-Q plot of residuals from a Mixture Cure Rate (MCR) model fitted via [MCRfit](#). Optionally, a simulation envelope can be included for Cox-Snell residuals.

**Usage**

```
qqMCR(
  object,
  type = c("cox-snell", "quantile"),
  envelope = FALSE,
  nsim = 100,
  censor = NULL,
  ...
)
```

**Arguments**

|                       |                                                                                                                                                                               |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>object</code>   | An object of class MCR, typically returned by <a href="#">MCRfit</a> .                                                                                                        |
| <code>type</code>     | Character. Type of residual to use in the QQ-plot. Options are "cox-snell" or "quantile". Defaults to "cox-snell".                                                            |
| <code>envelope</code> | Logical. Whether to add a simulation envelope to the QQ-plot. Default is FALSE.                                                                                               |
| <code>nsim</code>     | Integer. Number of simulations used to construct the envelope. Default is 100.                                                                                                |
| <code>censor</code>   | Logical vector or NULL. Censoring indicator used when simulating data for the envelope. Required only when <code>envelope = TRUE</code> and <code>type = "cox-snell"</code> . |
| <code>...</code>      | Additional arguments (currently ignored).                                                                                                                                     |

**Details**

The function generates QQ-plots of either Cox-Snell or quantile residuals. When `envelope = TRUE` and `type = "cox-snell"`, a simulation envelope is added using Monte Carlo replications.

**Value**

A QQ-plot is produced as a side effect. Nothing is returned.

**See Also**

[MCRfit](#), [residuals.MCR](#)

**Examples**

```
## Not run:
data(liver)
fit <- MCRfit(Surv(time, status) ~ age + medh + relapse + grade | sex + grade,
              data = liver, dist = "weibull", link = "logit")
qqMCR(fit, type = "cox-snell", envelope = TRUE, nsim = 50, censor = liver$status)

## End(Not run)
```

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|               |                                        |
|---------------|----------------------------------------|
| residuals.MCR | <i>Compute residuals for MCR model</i> |
|---------------|----------------------------------------|

---

### Description

This function computes Global Cox-Snell and randomized quantile residuals for objects of class MCR.

### Usage

```
## S3 method for class 'MCR'
residuals(object, type = c("cox-snell", "quantile"), ...)
```

### Arguments

|        |                                                                          |
|--------|--------------------------------------------------------------------------|
| object | An object of class MCR, typically returned from <a href="#">MCRfit</a> . |
| type   | Type of residual.                                                        |
| ...    | Additional arguments (not used).                                         |

### Value

A numeric vector of residuals.

### Examples

```
data(liver)
names(liver)

model <- MCRfit(
  survival::Surv(time, status) ~ age + medh + relapse + grade | sex + age + medh + grade,
  data = liver
)
summary(residuals(model,type="quantile"))
```

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|      |                                                                  |
|------|------------------------------------------------------------------|
| rMCM | <i>Generate Random Samples for Mixture Cure Rate (MCR) Model</i> |
|------|------------------------------------------------------------------|

---

### Description

Simulates survival data from a mixture cure rate model with covariates and user-defined link and latency distributions. Censoring is applied randomly.

**Usage**

```
rMCM(
  n,
  x,
  w,
  censor,
  alpha,
  beta,
  eta,
  dist = "weibull",
  link = "logit",
  tau = 1
)
```

**Arguments**

|               |                                                                                                                                      |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>n</b>      | Integer. Number of observations to simulate.                                                                                         |
| <b>x</b>      | Matrix or numeric. Covariate matrix for the latency component (must include intercept if needed).                                    |
| <b>w</b>      | Matrix or numeric. Covariate matrix for the cure component (no intercept assumed).                                                   |
| <b>censor</b> | Numeric. Maximum censoring time (uniformly distributed).                                                                             |
| <b>alpha</b>  | Numeric. Shape parameter for the survival distribution.                                                                              |
| <b>beta</b>   | Numeric vector. Coefficients for the latency part.                                                                                   |
| <b>eta</b>    | Numeric vector. Coefficients for the cure part.                                                                                      |
| <b>dist</b>   | Character. Distribution for the latency part. Options: "weibull", "lognormal", "loglogistic", "invgauss", "exponential", "rayleigh". |
| <b>link</b>   | Character. Link function for cure component. Options: "logit", "probit", "rplogit", "cauchit".                                       |
| <b>tau</b>    | Numeric. Tuning parameter for power link functions. Defaults to 1.                                                                   |

**Value**

A list with elements:

**time** Observed (possibly censored) survival time.

**status** Event indicator (1 = event, 0 = censored).

**x** Covariate matrix for the latency component.

**w** Covariate matrix for the cure component.

**pCcensur** Percentage of cured individuals.

**pUCensur** Percentage of censored cases among the uncured.

**Examples**

```
# Example: Simulating survival data using the inverse Gaussian distribution
library(EMGCR)

n <- 500
beta <- c(1, -1, -2)
```

```
eta <- c(0.5, -0.5)
alpha <- 1.5

p <- length(beta)
q <- length(eta)

set.seed(10)
X <- matrix(rnorm(n*(p-1),0,1),n,p-1)
X <- cbind(1,X)

set.seed(20)
W <- matrix(runif(n*q,-1,1),n,q)
W <- scale(W)

max_censoring <- 10

set.seed(1234)
sim_data <- rMCM(n=n, x = X, w = W,
               censor = max_censoring,
               beta = beta, eta = eta,
               alpha = alpha,
               link = "logit", dist = "invgauss", tau = 1)

names(sim_data)
head(sim_data)
attributes(sim_data)
attr(sim_data, "pCcensur")
attr(sim_data, "pUCCensur")
```



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